

TECHNICAL SPECIFICATIONS

Document: System Architecture & Technology Stack Overview

1. System Overview

The proposed application is a robust, offline-first desktop management system designed for high reliability and ease of use in environments with limited internet connectivity. It is built as a monolithic desktop application with an embedded local database, ensuring zero dependencies on external cloud infrastructure for core operations.

2. Core Technology Stack

Component	Technology	Description & Purpose
Programming Language	Python 3.12+	Core backend logic, data processing, and application structure.
GUI Framework	CustomTkinter	Modern, hardware-accelerated user interface supporting dark/light themes.
Database Engine	SQLite3	Embedded, serverless database for secure, local data storage. Zero configuration required.
Reporting Engine	ReportLab	Dynamic generation of A4 print-ready PDF reports and ledgers.
Image Processing	Pillow (PIL)	Efficient handling and rendering of UI images, icons, and diagrams.
Deployment	PyInstaller	Packages the application into a standalone Windows .exe file with all dependencies.

3. Architecture & Security Features

- **Role-Based Access Control (RBAC):** Hierarchical access (Admin, Manager, Officer) ensuring users only see what they are authorized to access.
- **Cryptographic Hashing:** All user passwords are encrypted using SHA-256 hashing algorithms; no plain-text credentials are saved.
- **Offline-First Design:** The system requires absolutely no active internet connection, minimizing external attack vectors and downtime.
- **Automated Data Redundancy:** Built-in automated SQLite database backups to prevent data loss, with one-click restore capabilities.
- **Referential Integrity:** Strict foreign key constraints and cascading operations within the database to prevent orphaned records.

4. Minimum Hardware Requirements

Hardware	Minimum Requirement
Operating System	Windows 10 / 11 (64-bit)
Processor (CPU)	Intel Core i3 or equivalent AMD processor
Memory (RAM)	4 GB RAM (8 GB recommended for optimal performance)
Storage	500 MB available disk space (SSD recommended)
Display Resolution	1366 x 768 minimum (1920 x 1080 recommended)